Single Factor Experiments

Completely Randomized Design, Randomized Complete Block Design and Latin Square Design: layout, ANOVA and CD.

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1 Single factor

Only **one factor** (the treatments) is varied; all other conditions are kept uniform.

CRD ↔ one-way ANOVA · RCBD ↔ two-way · Latin square ↔ three-way (rows, columns, treatments).

2 The three designs at a glance

	CRD	RCBD	Latin square (LSD)
Principles	Randomization, replication	+ local control (1 direction)	+ local control (2 directions)
Best for	Homogeneous units: lab, greenhouse	One known gradient, e.g. fertility	Two perpendicular gradients
Layout	Fully random	Blocks = replications; every treatment once per block	Each treatment once per row and column; v^2 plots
Error df	$v(r-1) = n-v$	$(v-1)(r-1)$	$(v-1)(v-2)$

For the same v and r, CRD has the **largest** error df, because nothing is set aside for blocking.

3 CRD

Model

$$y_{ij} = \mu + \tau_i + e_{ij}$$

Source	df
Treatments	$v-1$
Error	$n-v$
Total	$n-1$, with $n = rv$

Allows unequal replication. Rarely used in the field, where plots are seldom homogeneous.

4 RCBD

Model

$$y_{ij} = \mu + \tau_i + \beta_j + e_{ij}$$

Source	df
Blocks	$r-1$
Treatments	$v-1$
Error	$(v-1)(r-1)$

Blocks run **across** the gradient so that each block is uniform. Randomize independently within each block.

5 Latin Square Design

Model

$$y_{ijk} = \mu + \rho_i + \gamma_j + \tau_k + e_{ijk}$$

Source	df
Rows / Columns / Treatments	$v-1$ each
Error	$(v-1)(v-2)$
Total	v^2-1

CF $= G^2/v^2$. Suited to a modest number of treatments: too few leave little error df, too many need many homogeneous plots.

6 Critical difference (CD)

CRD (equal) & RCBD

$$t_{\alpha/2,\,edf}\sqrt{\frac{2\,MSE}{r}}$$

Latin square

$$t_{\alpha/2,\,edf}\sqrt{\frac{2\,MSE}{v}}$$

CRD (unequal r_i, r_j)

$$t_{\alpha/2,\,edf}\sqrt{MSE\left(\frac{1}{r_i}+\frac{1}{r_j}\right)}$$

Used only after a significant treatment F. Means sharing a letter are not significantly different.

7 How many replications?

A target of about **12 error df** is a practical planning guide, not a rule. Replication should also reflect the power and precision wanted, the expected variability and the available resources.

Correction factor = (grand total)² ÷ total number of observations. Treatment comparisons always use the **error mean square**.

! Common mistakes

- × Blocks laid **along** the gradient instead of across it.
- × Using RCBD error df for a CRD.
- × Applying CD when F is not significant.

? Fill in the blanks

1. A Latin square for 5 treatments needs ——— plots. [25]
2. A 5 × 5 Latin square has ——— error df. [12]
3. In an RCBD with 6 treatments and 4 blocks, each block has ——— plots. [6]
4. A pot trial in a uniform greenhouse is best laid out as a ———. [CRD]
5. If fertility falls from north to south, blocks should run ———. [east–west (across the gradient)]
6. For the same treatments and replications, ——— has the largest error df. [CRD]

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